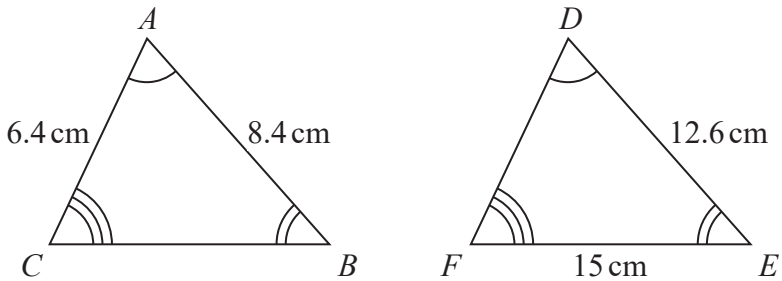
27 Triangle *ABC* and triangle *DEF* are similar.



(a) Work out the length of *DF*.

Match the angles: AB ↔ DE, AC ↔ DF, CB ↔ FE.

ABC	DEF
8.4	12.6
÷8.4	÷8.4
1	1.5
×6.4	×6.4
6.4	9.6

Scale factor = 12.6 ÷ 8.4 = 1.5

1 mark
Use corresponding sides

DF = 6.4 × 1.5 = 9.6 cm

1 mark
Correct corresponding length

9.6 cm

(2)

(b) Work out the length of *CB*.

ABC	DEF
1	1.5
×10	×10
10	15

CB = 15 ÷ 1.5 = 10 cm

1 mark
Reverse the scale factor

1 mark
Correct smaller length

10 cm

(2)

(Total for Question 27 is 4 marks)